

Project Synopsis: Analysis of Cost Overrun Factors in Construction Projects

1. Overview and Objective:

This document summarizes the analysis of a comprehensive dataset concerning the causes of cost overruns in construction projects, likely in a specific region (e.g., UAE, given the location data). The primary objective is to identify, categorize, and quantify the key factors that contribute to cost overruns, ultimately creating a risk assessment model.

2. Data Source:

The analysis is based on survey responses from 143 professionals (e.g., Project Managers, Engineers) working in the construction industry, representing clients, consultants, and contractors. The survey asked respondents to rate the impact of 44 predefined factors on cost overruns on a scale (likely 1 to 5).

3. Key Analysis Performed:

Factor Analysis: This was the core analytical technique. It reduced the 44 individual risk factors into a smaller, more manageable set of underlying "latent factors."

Factor Extraction: The analysis extracted 6 significant factors that explain a substantial portion (over 26%) of the common variance in the data. These factors were then rotated (Varimax) for easier interpretation.

Factor Interpretation: Based on the variables that loaded heavily onto each factor, the six factors were interpreted as: Construction execution / site management problems, Design and planning deficiencies, Financial and administrative delays, Scope changes / client changes, Supply and procurement issues, Contract management / external conditions.

Regression Analysis: A multiple regression model was built to predict the frequency of cost overruns. This model used the six factors, along with the respondent's qualifications and experience, as predictor variables.

Key Findings from Regression: Strongest Positive Predictors: Factor 5 (Supply and procurement issues) and Factor 6 (Contract management / external conditions) are the most significant drivers of higher cost overrun frequency.

Strongest Negative Predictor: Relevant working experience was a highly significant negative predictor, indicating that more experienced professionals report fewer cost overruns. Factor 4 (Scope changes/client changes) was also a significant negative predictor. Other Notable Predictors: The respondent's highest formal qualification was a marginally significant positive predictor. Factors 1, 2, and 3 were not statistically significant in this model.

Risk Score & Segmentation: A risk score was calculated for each respondent using the coefficients from the regression model, providing a single metric of their overall risk exposure. Respondents were segmented into risk groups (1 to 5). The analysis highlights the characteristics of the top 3 high-risk respondents (e.g., Project Managers with Master's degrees and 5-10 years of experience in Dubai) and the bottom 3 low-risk respondents (e.g., General Managers with Diplomas and over 15 years of experience in Sharjah).

4. Summary of Conclusions:

The primary causes of cost overruns in the surveyed construction projects are rooted in supply chain problems and contract management/external conditions. Experience is a critical mitigating factor. More experienced professionals are significantly less likely to report frequent cost overruns,

suggesting a strong role for tacit knowledge and practical expertise. The six-factor model provides a structured framework for understanding the multidimensional nature of project cost overruns, moving beyond a list of individual causes. The risk scores and segmentation offer a practical tool for identifying high-risk project profiles based